

1. Review midterm questions if necessary
2. Talk about $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, how $a + c \equiv b + d \pmod{n}$.

1 Proving non-conditional statements

1.1 If-and-only if

1. Review logic
2. Outline of the proof
3. Example: n is odd iff n^2 is odd
4. Example: $a, b \in \mathbb{Z}$. Then $a \equiv b \pmod{6}$ iff $a \equiv b \pmod{2}$ and $a \equiv b \pmod{3}$
5. Example: $a, b \in \mathbb{Z}$. Then $(a - 3)b^2$ is even iff a is odd or b is even.
6. Example: $a, b \in \mathbb{Z}$. Then $a + b$ is even iff a, b have the same parity.

1.2 Equivalent statements

Example: Suppose A is an $n \times n$ matrix and $\lambda \in \mathbb{R}$. TFAE

- λ is an eigenvalue of A
- $\det(A - \lambda I) = 0$
- The system of equations $Ax = \lambda x$ has a non-zero solution.
- The matrix $A - \lambda I$ is non-invertible

Example: Suppose $n \in \mathbb{Z}$. TFAE

- n is even
- $n = 2k$ for some $k \in \mathbb{Z}$
- etc...

2 Proofs involving sets

1. Recall definitions of union, intersection, difference, cartesian product and complement
2. Proving $a \in A$
3. Proving $A \subseteq B$
4. Proving $A = B$

Examples:

- $A = \{n \in \mathbb{N} : n \text{ has an odd number of distinct prime factors}\}$. Show that $7 \in A$, $15 \notin A$, and determine if 210 is an element of A .
- Let $A = \{n \in \mathbb{Z} : 8 \mid n\}$ and $B = \{n \in \mathbb{Z} : 4 \mid n\}$. Show that $A \subseteq B$.
- Let $A = \{n \in \mathbb{Z} : 3 \mid n, 5 \mid n\}$ and $B = \{n \in \mathbb{Z} : 15 \mid n\}$

More examples

Example 1. Let $A = \{(x, y) \in \mathbb{Z}^2 : x \equiv y \pmod{7}\}$. Show that $(12, -5) \in A$ and determine for which values of $n \in \mathbb{Z}$, $(5n + 4, 3 - 2n) \in A$.

Example 2. Let A, B be any sets. Then $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Example 3. $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$. is the converse true?

Example 4. For each $a \in \mathbb{R}$, let $A_a = \{(x, a(x^2 - 1)) \in \mathbb{R}^2 : x \in \mathbb{R}\}$. Prove that $\bigcap_{a \in \mathbb{R}} A_a = \{(-1, 0), (1, 0)\}$.

Example 5. Prove that $\{12a + 4b : a, b \in \mathbb{Z}\} = \{4n : n \in \mathbb{Z}\}$.